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Management apparatus for energy storage device and energy storage apparatus

Abstract

A BMU (management apparatus) of a secondary battery (energy storage device) includes a management part that operates by power supplied from the secondary battery, and a capacitor (backup power source) that supplies power to the management part when a voltage of the secondary battery decreases.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a management apparatus for an energy storage device and an energy storage apparatus.

BACKGROUND ART

(2) Conventionally, a management apparatus for an energy storage device, used for activating a drive source, is known (e.g., see Patent Document 1). The battery management apparatus described in Patent Document 1 manages an energy storage device used for starting (cranking) an engine of a vehicle. Generally, such a management apparatus operates by power supplied from the energy storage device.

PRIOR ART DOCUMENT

Patent Document

(3) Patent Document 1: Japanese Patent No. 6436201

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(4) As illustrated in FIG. 8A, a large discharge current (hereinafter, simply referred to as a large current) may temporarily flow through the energy storage device used for activating the drive source. For example, in the case of an energy storage device used for starting an engine of a vehicle, a large current temporarily flows from the energy storage device to a starter of the vehicle at the time of starting the engine.

(5) The capacity of the energy storage device used for activating the drive source varies.

Conventionally, sufficient studies have not been made on a problem in a case where a large current temporarily flows through an energy storage device having a small capacity.

Means for Solving the Problems

(6) In the present specification, a technique is disclosed in which, in a management apparatus that operates by power supplied from an energy storage device used for activation of a drive source, even when the capacity of the energy storage apparatus is small, an operation of the management apparatus is prevented from stopping due to instantaneous power interruption by a temporary large current.

(7) A management apparatus for an energy storage device used for activating a drive source according to one aspect of the present invention includes: a management part that operates by power supplied from the energy storage device; and a backup power source that temporarily supplies power to the management part when a voltage of the energy storage device decreases.

Advantages of the Invention

(8) In the management apparatus that operates by the power supplied from the energy storage device used for activating the drive source, it is possible to prevent the stop of the operation of the management apparatus due to instantaneous power interruption caused by the temporary large current even when the capacity of the energy storage apparatus is small.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view of a motorcycle.
- (2) FIG. 2 is a block diagram of a vehicle system.
- (3) FIG. 3 is an exploded perspective view of a battery.
- (4) FIG. 4 is a plan view of a secondary battery.
- (5) FIG. 5 is a cross-sectional view taken along line A-A in FIG. 4.
- (6) FIG. 6 is a block diagram of a battery.
- (7) FIG. 7 is a circuit diagram of an alternator, an energy storage device, a management part, and a backup power source.
- (8) FIG. 8A is a graph illustrating a change in the discharge current of the energy storage device.
- (9) FIG. 8B is a graph illustrating a change in the voltage of the energy storage device.

MODE FOR CARRYING OUT THE INVENTION

Summary of Present Embodiment

- (10) A management apparatus for an energy storage device used for activating a drive source includes: a management part that operates by power supplied from the energy storage device; and a backup power source that temporarily supplies power to the management part when a voltage of the energy storage device decreases.
- (11) The inventor of the present application has found that, when a large discharge current (large current) temporarily flows in an energy storage device having a small capacity, there is a higher possibility than in an energy storage device having a large capacity that the operation of the management apparatus stops due to instantaneous power interruption (blackout) in which the voltage of the energy storage device temporarily becomes lower the operating voltage of the management part. Hereinafter, a specific description will be given with reference to FIG. 8B. In the following description, it is assumed that the operating voltage of the management apparatus is 5 V.
- (12) When the capacity of the energy storage device is large, a decrease in voltage is small even when a large current flows. Therefore, as indicated by a dotted line **21** in FIG. 8B, even when a large current temporarily flows, the voltage is unlikely to decrease to less than the operating voltage (5 V) of the management part. In contrast, as indicated by a solid line **22**, when the capacity of the energy storage device is small, the voltage greatly decreases even with the current of the same magnitude. Hence, it becomes more likely that the voltage decreases to less than 5 V. In other words, when the capacity of the energy storage device is small, there is a high possibility that instantaneous power interruption occurs.
- (13) According to the management apparatus, since the backup power source is provided, when the voltage of the energy storage device becomes lower than the voltage of the backup power source, power is supplied from the backup power source to the management part. This can prevent instantaneous power interruption. Therefore, in the management apparatus that operates by the power supplied from the energy storage device used for activating the drive source, it is possible to

prevent the stop of the operation of the management apparatus due to instantaneous power interruption caused by the temporary large current even when the capacity of the energy storage apparatus is small.

(14) When the temperature of the energy storage device is low, instantaneous power interruption is likely to occur due to an increase in internal resistance. According to the management apparatus, instantaneous power interruption can be prevented, so that the management apparatus is particularly useful in the case of being used at a low temperature when the internal resistance of the energy storage device is high or in a state where the internal resistance of the energy storage device has greatly increased.

(15) The energy storage device may be used for starting an engine of a motorcycle.

(16) In general, the magnitude of the charge-discharge current of the energy storage device is represented by C-rate. The C-rate is obtained by defining, as 1 C, a magnitude of a current flowing when an energy storage device in a state of charge (SOC) of 100% is discharged to 0% in one hour (alternatively, a magnitude of a current flowing in a case where the energy storage device in a SOC of 0% is charged to 100% in one hour). For example, when the energy storage device is discharged from the SOC of 100% to 0% in 30 minutes, the C-rate is 2 C.

(17) When charge capacities of energy storage devices are different, C-rates are different even when current values of charge-discharge currents are the same. For example, it is assumed that there are an energy storage device B1 having a charge capacity of 100 Ah [ampere hour] and an energy storage device B2 having a charge capacity of 200 Ah, and the current value of the discharge current of each of these energy storage devices is 100 A [ampere]. In this case, the SOC of the energy storage device B1 becomes 0% in one hour, and hence the C-rate becomes 1 C. In contrast, the SOC of the energy storage device B2 becomes 0% in two hours, and hence the C-rate becomes 0.5 C. As described above, when the current value of the charge-discharge current is the same, the smaller the charge capacity is, the larger the C-rate becomes. In the following description, the C-rate being high is referred to as a high rate, and the C-rate being low is referred to as a low rate.

(18) Normally, an energy storage device used for starting an engine of a motorcycle (an energy storage device for a two-wheeled vehicle) has a smaller charge capacity and a smaller charge-discharge current than an energy storage device used for starting an engine of a four-wheeled automobile (an energy storage device for a four-wheeled vehicle). However, the difference between the magnitudes of the charge-discharge currents is not as large as the difference between the charge capacities. For example, the charge capacity of the energy storage device for a two-wheeled vehicle is about one-third of the charge capacity of the energy storage device for a four-wheeled vehicle, and the magnitude of the discharge current of the energy storage device for a two-wheeled vehicle is about four-fifths of the discharge current of the energy storage device for a four-wheeled vehicle. Therefore, in general, the energy storage device for a two-wheeled vehicle is charged and discharged at a higher rate than the energy storage device for a four-wheeled vehicle.

(19) In the case of discharge at a low rate, the voltage drop of the energy storage device is small as compared to the case of discharge at a high rate, so that there is a less possibility of instantaneous power interruption. For this reason, measures for preventing instantaneous power interruption have not been taken in the energy storage device that discharges at a low rate.

(20) In contrast, in the case of discharge at a high rate, a decrease in voltage of the energy storage device is large as compared to the case of discharge at a low rate. The inventor of the present application focused on this point and found that the energy storage device for a two-wheeled vehicle discharged at a high rate is highly likely to cause instantaneous power interruption.

(21) According to the management apparatus, the backup power source is provided in the management apparatus of the energy storage device for a two-wheeled vehicle in which instantaneous power interruption easily occurs, so that the instantaneous power interruption can be prevented in the management apparatus of the energy storage apparatus for a two-wheeled vehicle.

(22) In general, the management apparatus of the energy storage device for a two-wheeled vehicle has low power consumption as compared to the management apparatus of the energy storage device for a four-wheeled vehicle, and hence the capacity of the backup power source may also be small. Therefore, there is also an advantage that instantaneous power interruption can be prevented at a low cost as compared to a case where the backup power source is provided in the management apparatus of the energy storage device for a four-wheeled vehicle.

(23) The backup power source may temporarily supply power to the management part when the voltage of the energy storage device decreases.

(24) In order to prevent instantaneous power interruption, power only has to be temporarily supplied to the management part. In this case, when a backup power source capable of supplying power for a certain long time is used, the backup power source becomes excessive, and the manufacturing cost of the management apparatus increases. According to the management apparatus, the backup power source temporarily supplies power to the management part, so that the manufacturing cost of the management apparatus can be held down as compared to the case of using the backup power source that can supply power for a certain long time.

(25) The times during which the backup power source temporarily supplies power to the management part can be appropriately selected, but when the times during which the backup power source supplies power are long, it is necessary to increase the capacity of the backup power source. Thus, the time during which power is supplied is desirably within ten seconds, and more desirably within five seconds. When the duration of the instantaneous power interruption is less than one second, the times during which the backup power source supplies power may be within one second.

(26) The backup power source may be a capacitor charged by the energy storage device.

(27) Generally, the cost of the capacitor is low, and it is thus possible to achieve a backup power source that temporarily supplies power to the management part.

(28) Assuming that the backup power source requires the control of the management part, when the voltage of the energy storage device decreases to, for example, a voltage close to 5 V, the management part needs to supply power to the backup power source by software control. However, in that case, the processing of the management part becomes complicated, and the response speed also becomes slow due to the software control. In contrast, when the voltage of the energy storage device becomes lower than the voltage of the capacitor, the capacitor automatically supplies power to the management part, so that the software control is unnecessary. Therefore, there is also an advantage that the processing of the management part is simplified. When the capacitor is used, instantaneous power interruption can be handled only by hardware, and hence there is also an advantage that instantaneous power interruption can be handled quickly as compared to the software control.

(29) When the capacitor is used, there is also an advantage that noise can be removed at the same time.

(30) The management apparatus may include a circuit breaker connected in series with the energy storage device, and a current sensor that detects a current value of a current flowing through the energy storage device, and the management part may turn off the circuit breaker when a current value equal to or larger than a predetermined value is detected by the current sensor.

(31) When an external short circuit occurs, a large current (a current equal to or larger than a predetermined value) flows through the energy storage device. According to the management apparatus, when a large current flows, the energy storage device can be protected from the large current by the management part turning off the circuit breaker. However, when the voltage of the energy storage device becomes lower than the operating voltage of the management part due to the large current, the operation of the management part stops without the backup power source, so that the circuit breaker cannot be turned off.

(32) According to the management apparatus, when the voltage of the energy storage device becomes lower than the voltage of the backup power source, the power is supplied from the backup

power source to the management part. Thus, the management part operated by the power supplied from the backup power source turns off the circuit breaker, whereby the energy storage device can be protected from coming into the state of not being able to control the discharge of a large current due to an external short circuit.

(33) A regulator that lowers a voltage of the energy storage device to the operating voltage of the management part may be provided between the energy storage device and the management part, and the backup power source may be provided on a power line branching from a power line connecting the regulator and the management part.

(34) When the backup power source is provided, the backup power source can be provided between the energy storage device and the regulator, but in this case, a backup power source having a large capacity capable of storing the same voltage as the voltage of the energy storage device is required. In contrast, when the backup power source is provided between the regulator and the management part, the voltage is lowered by the regulator, so that the capacity of the backup power source can be reduced as compared to the case where the backup power source is provided between the energy storage device and the regulator. Therefore, instantaneous power interruption can be prevented at a lower cost.

(35) The invention disclosed by the present specification can be achieved in various aspects such as an apparatus, a method, a computer program for achieving the apparatus or the method, and a recording medium on which the computer program is recorded.

First Embodiment

(36) A first embodiment will be described with reference to FIGS. 1 to 8.

(37) As illustrated in FIG. 1, a battery **50** (energy storage apparatus) according to the first embodiment is a battery for a two-wheeled vehicle mounted on a motorcycle **10**.

(38) As illustrated in FIG. 2, a starter **10A**, an alternator **10B**, and auxiliary equipment **10C** (headlights, air conditioners, audios, etc.), which are mounted on the motorcycle **10**, are connected to the battery **50**. The battery **50** is an engine starting battery that supplies power to the starter **10A** to start an engine (an example of a drive source). The battery **50** is charged by the alternator **10B** during engine operation.

(39) Power is supplied from the alternator **10B** to the auxiliary equipment **10C** during the engine operation of the motorcycle **10**. Thus, the battery **50** does not supply power to the auxiliary equipment **10C** while the engine is operating, but when the auxiliary equipment **10C** is used while the engine is stopped, the battery **50** also supplies power to the auxiliary equipment **10C**.

(40) In general, a battery for a two-wheeled vehicle does not have a function of communicating with an electronic control unit (ECU) of the motorcycle **10**. The battery **50** for a two-wheeled vehicle according to the first embodiment also does not have a function of communicating with the ECU.

(41) (1) As illustrated in FIG. 3, the battery **50** includes an assembled battery **60**, a circuit board unit **65**, and a housing **71**.

(42) The housing **71** includes a body **73** made of a synthetic resin material and a lid body **74**. The body **73** has a bottomed cylindrical shape. The body **73** includes a bottom surface portion **75** and four side surface portions **76**. An upper opening **77** is formed in an upper-end portion by the four side surface portions **76**.

(43) The housing **71** houses the assembled battery **60** and a circuit board unit **65**. The assembled battery **60** includes 12 secondary batteries **62** (an example of an energy storage device). The secondary battery **62** is, for example, a lithium ion battery. The twelve secondary batteries **62** are connected with three in parallel and four in series. The circuit board unit **65** includes a circuit board **100** and an electronic component mounted on the circuit board **100** and is disposed on the assembled battery **60**.

(44) The lid body **74** closes the upper opening **77** of the body **73**. An outer peripheral wall **78** is provided around the lid body **74**. The lid body **74** has a protrusion **79** in a substantially T-shape in a

plan view. A positive external terminal **51** is fixed to one corner of the front portion of the lid body **74**, and a negative external terminal **52** is fixed to the other corner.

(45) As illustrated in FIGS. **4** and **5**, the secondary battery **62** houses an electrode assembly **83** together with a nonaqueous electrolyte in a case **82** having a rectangular parallelepiped shape. The case **82** has a case body **84** and a lid **85** for closing an opening at the top of the case body **84**.

(46) Although not shown in detail, the electrode assembly **83** has a separator, made of a porous resin film, disposed between a negative electrode element formed by applying an active material to a substrate made of copper foil and a positive electrode element formed by applying an active material to a substrate made of aluminum foil. These are all belt-shaped, and are wound in a flat shape so as to be able to be housed in the case body **84** in a state where the negative electrode element and the positive electrode element are displaced from each other on the opposite sides in the width direction with respect to the separator.

(47) The positive electrode element is connected to a positive electrode terminal **87** via a positive current collector **86**, and the negative electrode element is connected to a negative electrode terminal **89** via a negative current collector **88**. Each of the positive current collector **86** and the negative current collector **88** is made up of a plate-shaped base **90** and legs **91** extending from the base **90**. A through hole is formed in the base **90**. The leg **91** is connected to the positive electrode element or the negative electrode element. Each of the positive electrode terminal **87** and the negative electrode terminal **89** is made up of a terminal body portion **92** and a shaft **93** protruding downward from the center portion of the lower surface of the terminal body portion **92**. Among those, the terminal body portion **92** and the shaft **93** of the positive electrode terminal **87** are integrally formed of aluminum (single material). The negative electrode terminal **89** has the terminal body portion **92** made of aluminum and the shaft **93** made of copper and is assembled with these. The terminal body portions **92** of the positive electrode terminal **87** and the negative electrode terminal **89** are disposed at both ends of the lid **85** via gaskets **94** made of an insulating material and are exposed outward from the gaskets **94**.

(48) The lid **85** has a pressure release valve **95**. As illustrated in FIG. **2**, the pressure release valve **95** is located between the positive electrode terminal **87** and the negative electrode terminal **89**. The pressure release valve **95** is opened when the internal pressure of the case **82** exceeds a limit value to lower the internal pressure of the case **82**.

(2) Electrical Configuration of Battery

(49) As illustrated in FIG. **6**, the battery **50** includes an assembled battery **60** and a battery management unit (BMU) **101** that manages the assembled battery **60**. The BMU **101** is an example of a management apparatus.

(50) The assembled battery **60** includes a plurality of secondary batteries **62**. The twelve secondary batteries **62** are connected with three in parallel and four in series. In FIG. **6**, three secondary batteries **62** connected in parallel are represented by one battery symbol. The battery **50** is rated at 12 V.

(51) A power line **70P** is a power line for connecting the positive external terminal **51** and the positive electrode of the assembled battery **60**. A power line **70N** is a power line for connecting the negative external terminal **52** to the negative electrode of the assembled battery **60**. The negative electrode of the assembled battery **60** is connected to a signal ground **G1**. The assembled battery **60** has the signal ground **G1** as a reference potential. The negative external terminal **52** is connected to a body ground **G2**. The body ground **G2** is the body of the motorcycle **10**. The body ground **G2** is the reference potential of the motorcycle **10**.

(52) The BMU **101** includes a current sensor **53**, a voltage sensor **110**, a temperature sensor **111**, a circuit breaker **55**, a management part **130**, a diode **40** (rectifier device), a capacitor **41** (an example of a backup power source), and a regulator **42**. The assembled battery **60**, the current sensor **53**, and the circuit breaker **55** are connected in series via the power line **70P** and the power line **70N**. The circuit breaker **55**, the current sensor **53**, the voltage sensor **110**, the management part **130**, the

diode **40**, the capacitor **41**, and the regulator **42** are mounted on the circuit board **100**, and the signal ground **G1** of the circuit board **100** is set as a reference potential (operation reference).

(53) The current sensor **53** is located on the negative electrode side of the assembled battery **60** and is provided on the power line **70N** on the negative electrode side. The current sensor **53** detects a current I of the assembled battery **60**.

(54) The voltage sensor **110** detects a voltage V of each secondary battery **62** and the total voltage of the assembled battery **60**. The total voltage of the assembled battery **60** is the total voltage of the four secondary batteries **62**.

(55) The temperature sensor **111** is provided in any one or two of the secondary batteries **62**, detects the temperature of the secondary battery **62**, and outputs the temperature to the management part **130**.

(56) The circuit breaker **55** is located at the negative electrode of the assembled battery **60** and is provided in the power line **70N** of the negative electrode. The circuit breaker **55** includes a charging field-effect transistor (FET) **55A** and a discharging FET **55B**. The charging FET **55A** and the discharging FET **55B** are power semiconductor switches, more specifically, N-channel field-effect transistors (FETs). Sources S of the charging FET **55A** and discharging FET **55B** are reference terminals. Gates G of the charging FET **55A** and the discharging FET **55B** are control terminals. Drains D of the charging FET **55A** and discharging FET **55B** are connection terminals.

(57) In the charging FET **55A**, the source S is connected to the negative electrode of the assembled battery **60**. The source S of the discharging FET **55B** is connected to the negative external terminal **52**. The charging FET **55A** and the discharging FET **55B** are connected back-to-back by the drains D being connected to each other.

(58) The charging FET **55A** includes a parasitic diode **56A**. The forward direction of the parasitic diode **56A** is the same as the discharge direction. The discharging FET **55B** includes a parasitic diode **56B**. The forward direction of the parasitic diode **56B** is the same as the charging direction.

(59) Since the source S of the discharging FET **55B** is connected to the negative external terminal **52**, the body ground **G2** is a reference potential. In the charging FET **55A**, the source S is connected to the negative electrode of the assembled battery **60**. Since the negative electrode of the assembled battery **60** is connected to the signal ground **G1** of the circuit board **100**, the signal ground **G1** is a reference potential in the charging FET **55A**.

(60) The charging FET **55A** is turned on by an H-level voltage being applied to the gate G and is turned off by an L-level voltage being applied to the gate G . The same applies to the discharging FET **55B**.

(61) The management part **130** is operated by power supplied from the secondary battery **62** and includes a central processing unit (CPU) **131**, a read-only memory (ROM) **132**, and a random-access memory (RAM) **133**. The operating voltage of the management part **130** is 5 V. The management part **130** manages the battery **50** based on the outputs of the voltage sensor **110**, the current sensor **53**, and the temperature sensor **111**. In a normal state, the management part **130** applies an H-level voltage to each of the gate G of the charging FET **55A** and the gate G of the discharging FET **55B** to turn on the charging FET **55A** and the discharging FET **55B**. When both the charging FET **55A** and the discharging FET **55B** are on, the assembled battery **60** can perform both charge and discharge.

(62) The regulator **42** is a circuit that steps down a voltage of 12 V supplied from the secondary battery **62** to 5 V that is the operating voltage of the management part **130**. The regulator **42** is provided in a power line **120** that supplies the power of the secondary battery **62** to the management part **130**. The power line **120** branches from the power line **70P**. When the operating voltage of the management part **130** is 12 V or when the management part **130** itself includes a regulator, the regulator **42** is unnecessary.

(63) The diode **40** is provided between the regulator **42** and the assembled battery **60** in the power line **120**. A current flows through the diode **40** in a direction from the secondary battery **62** toward

the management part **130**. The diode **40** prevents the voltage of the capacitor **41** from decreasing when the voltage of the assembled battery **60** decreases.

(64) The capacitor **41** is provided in a power line **121** branching from between the diode **40** and the regulator **42** in the power line **120**. The capacitor **41** is charged with 12 V by the assembled battery **60**, and when the total voltage of the assembled battery **60** becomes lower than the voltage of the capacitor **41**, power of 5 V or more is temporarily supplied to the management part **130**.

(65) The time during which the capacitor **41** temporarily supplies power to the management part **130** can be selected as appropriate, but when the time during which the power is supplied is long, it is necessary to increase the capacity of the capacitor **41**. Thus, the time during which power is supplied is desirably within ten seconds, and more desirably within five seconds. When the time during which instantaneous power interruption continues is less than one second, the time during which the capacitor **41** supplies power may be within one second.

(3) Processing Executed by Management Part

(66) Of the processing executed by the management part **130**, SOC estimation processing, overcharge-overdischarge protection processing, and overcurrent protection processing will be described.

(67) (3-1) SOC Estimation Processing

(68) The SOC estimation processing is processing of estimating the state of charge of the secondary battery **62** by the current integration method. The current integration method is a method in which the amount of power flowing into and out of the secondary battery **62** is measured by measuring the charge-discharge current of the secondary battery **62** at predetermined time intervals with the current sensor **53** and is then added to or subtracted from the initial capacity to estimate the SOC.

(69) The current integration method has the advantage that the SOC can be estimated even when the secondary battery **62** is in use, but the current sensor **53** always measures the current and integrates the charge-discharge power amount, so that the measurement error of the current sensor **53** may accumulate and become increasingly inaccurate. Therefore, the management part **130** may reset the SOC to be estimated by the current integration method based on the open-circuit voltage (OCV) of the secondary battery **62**.

(70) Specifically, there is a relatively accurate correlation between the OCV and the SOC, and hence the SOC may be estimated from the OCV, and the SOC estimated by the current integration method may be reset to the SOC estimated from the OCV.

(71) The OCV is not limited to a voltage in a state where the circuit is open. For example, the OCV may be a voltage when the current value of the current flowing through the secondary battery **62** is less than a minute reference value.

(72) (3-2) Overcharge-Overdischarge Protection Processing

(73) The overcharge-overdischarge protection processing is processing of protecting the secondary battery **62** from overcharge by turning off the charging FET **55A** when the SOC becomes equal to or larger than a predetermined value and is processing of protecting the secondary battery **62** from overdischarge by turning off the discharging FET **55B** when the SOC becomes equal to or smaller than a predetermined lower limit value.

(74) (3-3) Overcurrent Protection Processing

(75) The overcurrent protection processing is processing of protecting the secondary battery **62** from a large current by turning off the discharging FET **55B** when a large current flows through the secondary battery **62** due to an external short circuit. Specifically, the management part **130** detects a current value at predetermined time intervals with the current sensor **53** and turns off the discharging FET **55B** when a current value equal to or larger than a predetermined value is detected. The predetermined value is a value larger than a current value of a large current flowing at the time of engine start.

(4) Operation of Capacitor

(76) The operation of the capacitor **41** will be described with reference to FIG. 7. With the

assembled battery **60** being charged and discharged at a high rate, a large current may temporarily flow at the time of starting the engine, so that the total voltage may decrease to less than 5 V. When the total voltage of the assembled battery **60** decreases to less than 5 V, the total voltage of the assembled battery **60** becomes lower than the voltage of the capacitor **41**, and hence power is temporarily supplied from the capacitor **41** to the management part **130**.

(77) Since it is temporary that the total voltage of the assembled battery **60** decreases to less than 5 V at the time of starting the engine, when the voltage of the assembled battery **60** recovers thereafter, power is supplied from the assembled battery **60** to the management part **130**. Therefore, the management part **130** continues to operate even when the total voltage of the assembled battery **60** temporarily decreases to less than 5 V at the time of starting the engine. In other words, the SOC estimation processing and the overcharge-overdischarge protection processing described above are executed continuously.

(78) With the assembled battery **60** being charged and discharged at a high rate, even when an external short circuit occurs, the total voltage of the assembled battery **60** may decrease to less than 5 V due to a large current flowing. Even when the total voltage of the assembled battery **60** decreases to less than 5 V due to an external short circuit, power is temporarily supplied from the capacitor **41** to the management part **130**.

(79) The reason why power is supplied from the capacitor **41** to the management part **130** when an external short circuit occurs is that the management part **130** is required to be temporarily utilized so as to execute the overcurrent protection processing described above. Specifically, when a large current flows due to an external short circuit, the current value becomes equal to or larger than a predetermined value, so that the management part **130** operated by the power supplied from the capacitor **41** detects that the current value is equal to or larger than the predetermined value, and turns off the discharging FET **55B**. As a result, the secondary battery **62** is protected from a large current due to an external short circuit. Thereafter, when power is no longer supplied from the capacitor **41**, the management part **130** stops operating.

(5) Effects of Embodiment

(80) According to the BMU **101**, the backup power source (capacitor **41**) is provided, and when the voltage of the assembled battery **60** becomes lower than the voltage of the backup power source, power is supplied from the backup power source to the management part **130**. This can prevent instantaneous power interruption. Thus, in the BMU **101** operated by the power supplied from the two-wheeled-bicycle secondary battery **62**, even when the capacity of the secondary battery **62** is small, it is possible to prevent the stop of the operation of the BMU **101** due to the instantaneous power interruption by the temporary large current.

(81) When the temperature of the secondary battery **62** is low, instantaneous power interruption is likely to occur due to an increase in internal resistance. According to the BMU **101**, instantaneous power interruption can be prevented, so that the BMU **101** is particularly useful in the case of being used at a low temperature when the internal resistance of the secondary battery **62** is high or in a state where the internal resistance of the secondary battery **62** has greatly deteriorated.

(82) According to the BMU **101**, since the backup power source is provided in the BMU **101** of the secondary battery **62** for a two-wheeled vehicle in which instantaneous power interruption easily occurs, the instantaneous power interruption can be prevented in the secondary battery **62** for a two-wheeled vehicle.

(83) In general, the BMU **101** of the secondary battery **62** for a two-wheeled vehicle has a low operating voltage as compared to the BMU of the secondary battery **62** for a four-wheeled vehicle, and hence the capacity of the backup power source may also be small. Therefore, there is also an advantage that instantaneous power interruption can be prevented at a low cost as compared to a case where the backup power source is provided in the BMU of the secondary battery **62** for a four-wheeled vehicle.

(84) According to the BMU **101**, the backup power source temporarily supplies power to the

management part **130**, so that the manufacturing cost of the BMU **101** can be held down as compared to the case of using the backup power source that can supply power for a certain long time.

(85) According to the BMU **101**, the backup power source is a capacitor **41** charged by the secondary battery **62**. Generally, the cost of the capacitor is low, and it is thus possible to achieve a backup power source that temporarily supplies power to the management part.

(86) When the voltage of the energy storage device becomes lower than the voltage of the capacitor **41**, the capacitor **41** automatically supplies power to the management part **130**, so that the software control is unnecessary. Therefore, there is also an advantage that the processing of the management part **130** is simplified. When the capacitor **41** is used, instantaneous power interruption can be handled only by hardware, and hence there is also an advantage that instantaneous power interruption can be handled quickly as compared to the software control.

(87) When the capacitor **41** is used, there is also an advantage that noise can be removed at the same time.

(88) According to the BMU **101**, even when the voltage of the assembled battery **60** decreases due to the external short circuit, the management part **130** supplied with power from the capacitor **41** turns off the discharging FET **55B**, whereby the secondary battery **62** can be protected from being in the state of not being able to control the discharge of a large current due to an external short circuit.

Other Embodiments

(89) The technique disclosed in the present specification is not limited to the embodiment described with reference to the above description and drawings, and for example, such embodiments as follows are also included in the technical scope disclosed in the present specification.

(90) (1) In the above embodiment, the capacitor **41** has been described as an example of the backup power source, but the backup power source is not limited thereto. For example, the backup power source may be a primary battery such as a button battery (an example of a backup power source that can supply power for a certain long time).

(2) In the above embodiment, the case where the capacitor **41** is provided between the secondary battery **62** and the regulator **42** has been described as an example, but the capacitor **41** may be provided between the regulator **42** and the management part **130**. When the backup power source is provided between the regulator **42** and the management part **130**, the voltage is lowered by the regulator **42**, so that the capacity of the capacitor **41** can be reduced as compared to the case where the capacitor **41** is provided between the secondary battery **62** and the regulator **42**. Therefore, instantaneous power interruption can be prevented at a lower cost.

(3) In the above embodiment, the secondary battery **62** for a two-wheeled vehicle has been described as an example of the secondary battery used for activating the drive source, but the secondary battery used for starting the drive source is not limited thereto. For example, an electric vehicle, a hybrid vehicle, or the like includes a driving battery for supplying power to an electric motor that is a drive source of the vehicle. In general, a driving battery is started by power supplied from an external secondary battery and supplies power to an electric motor. The secondary battery used for starting the drive source may be used as the external secondary battery described above.

(4) In the above embodiment, the secondary battery **62** has been described as an example of the energy storage device, but the energy storage device is not limited thereto. For example, the energy storage device may be a capacitor accompanied by an electrochemical reaction.

DESCRIPTION OF REFERENCE SIGNS

(91) **10**: motorcycle **41**: capacitor (example of backup power source) **42**: regulator **53**: current sensor **55**: circuit breaker **62**: secondary battery (example of energy storage device) **101**: BMU (example of management apparatus) **130**: management part

Claims

1. A management apparatus for an energy storage device used for activating a drive source, the management apparatus comprising: a management part that operates by power supplied from the energy storage device; a backup power source that supplies power to the management part when a voltage of the energy storage device decreases; a circuit breaker connected in series with the energy storage device; and a current sensor that detects a current value of a current flowing through the energy storage device, wherein the management part turns off the circuit breaker when a current value equal to or larger than a predetermined value is detected by the current sensor.
2. The management apparatus according to claim 1, wherein the energy storage device is for starting an engine of a motorcycle.
3. The management apparatus according to claim 1, wherein the backup power source temporarily supplies power to the management part when a voltage of the energy storage device decreases.
4. The management apparatus according to claim 3, wherein the backup power source comprises a capacitor charged by the energy storage device.
5. The management apparatus according to claim 1, wherein the management part turns off the circuit breaker when a current value equal to or larger than the predetermined value is detected by the current sensor due to an external short circuit.
6. A management apparatus for an energy storage device used for activating a drive source, the management apparatus comprising: a management part that operates by power supplied from the energy storage device; a backup power source that supplies power to the management part when a voltage of the energy storage device decreases; and a regulator that lowers a voltage of the energy storage device to an operating voltage of the management part, the regulator provided between the energy storage device and the management part, wherein the backup power source is provided on a power line branching from a power line connecting the regulator and the management part.
7. An energy storage apparatus comprising: an energy storage device; and a management apparatus, wherein the management apparatus includes a management part that operates by power supplied from the energy storage device, and a backup power source that supplies power to the management part when a voltage of the energy storage device decreases, the energy storage apparatus is rated at 12 volts, the management apparatus includes a circuit breaker connected in series with the energy storage device, and a current sensor that detects a current value of a current flowing through the energy storage device, and the management part turns off the circuit breaker when a current value equal to or larger than a predetermined value is detected by the current sensor.
8. The energy storage apparatus according to claim 7, wherein an operating voltage of the management part included in the management apparatus is 5 V.
9. The energy storage apparatus according to claim 7, wherein the management part turns off the circuit breaker when a current value equal to or larger than the predetermined value is detected by the current sensor due to an external short circuit.
10. The energy storage apparatus according to claim 7, wherein the energy storage apparatus is for a motorcycle.
11. The energy storage apparatus according to claim 7, wherein the energy storage apparatus is used for activating a driving battery of an automobile.
12. The energy storage apparatus according to claim 7, further comprising a housing that houses the energy storage device and the management apparatus.
13. The energy storage apparatus according to claim 12, wherein a positive external terminal connected to a positive electrode of the energy storage device by a power line and a negative external terminal connected to a negative electrode of the energy storage device by the power line are fixed to the housing.
14. The management apparatus according to claim 1, wherein the backup power source temporarily

supplies power for a predetermined time to the management part when a voltage of the energy storage device decreases.

15. The management apparatus according to claim 1, wherein the backup power source exclusively supplies power to the management part when a voltage of the energy storage device decreases to a predetermined level.

16. The energy storage apparatus according to claim 7, wherein the backup power source exclusively supplies power to the management part when a voltage of the energy storage device decreases to a predetermined level.

17. An energy storage apparatus comprising: an energy storage device; and a management apparatus, wherein the management apparatus includes a management part that operates by power supplied from the energy storage device, and a backup power source that supplies power to the management part when a voltage of the energy storage device decreases, the energy storage apparatus is rated at 12 volts, a regulator that lowers a voltage of the energy storage device to an operating voltage of the management part is provided between the energy storage device and the management part, and the backup power source is provided on a power line branching from a power line connecting the regulator and the management part.
